

COMPARISON OF PATELLOFEMORAL JOINT LOADING ESTIMATES ACROSS MODELLING METHODS

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INTRODUCTION

Patellofemoral joint reaction force (PFJRF) and stress (PFJS) are commonly used to evaluate patellofemoral joint (PFJ) loading during movement. Musculoskeletal modelling methods can produce different absolute PFJ loading estimates, but the extent to which these differences change interpretation across tasks, populations, and conditions remains unclear. This study compared PFJRF and PFJS impulse estimates derived using different modelling methods and tested whether method choice changed conclusions regarding loading differences between running speeds, populations, and tasks.

METHODS

PFJ loading was analysed across three datasets: (1) running at different speeds, (2) runners with and without patellofemoral pain, and (3) squatting. PFJRF and PFJS were estimated using inverse dynamics (ID), static optimisation (SO), and dynamic optimisation (DO). PFJRF and PFJS impulse were calculated for each participant and method. Linear mixed effects models evaluated differences in impulse outcomes between methods and tested whether these differences varied by speed or participant group.

RESULTS AND DISCUSSION

Modelling method significantly affected PFJRF and PFJS impulse across all three datasets. ID generally

produced the lowest PFJ loading estimates, whereas DO produced the highest estimates. In the running speed dataset, PFJRF and PFJS impulse did not differ significantly between speeds, and method by speed interactions were not significant. In the patellofemoral pain dataset, PFJRF and PFJS impulse did not differ significantly between groups, and method by group interactions were not significant. In the squatting dataset, method effects were significant for both PFJRF and PFJS impulse

CONCLUSIONS

Musculoskeletal modelling method significantly influenced absolute PFJRF and PFJS impulse. PFJRF and PFJS impulse showed no significant speed or population differences, and these comparisons were robust to modelling method choice. Methodological consistency is therefore particularly important when comparing absolute PFJ loading estimates, but may be less critical when drawing conclusions about differences in PFJRF and PFJS impulse between speeds, populations.

REFERENCES

- [1] Hart HF et al. Br J Sports Med **56**: 521-30, 2022.
- [2] Scherpereel K et al. Sci Data **10**: 924, 2023.
- [3] Fukuchi RK et al. PeerJ **5**: e3298, 2017.
- [4] Fox AS. R Soc Open Sci **11**: 231909, 2024.
- [5] Kernozek TW et al. Gait Posture **42**: 218-22, 2015.

Table 1: Summary of statistical effects on PFJ loading outcomes across datasets

Dataset	Outcome	Method effect	Condition/Group effect	Interaction
Running speed	PFJRF impulse	$p < 0.001$	Speed: NS	Method $\times$ Speed: NS
	PFJS impulse	$p < 0.001$	Speed: NS	Method $\times$ Speed: NS
PFP vs healthy runners	PFJRF impulse	$p < 0.001$	Group: NS	Group $\times$ Method: NS
	PFJS impulse	$p < 0.001$	Group: NS	Group $\times$ Method: NS
Squatting	PFJRF impulse	$p < 0.001$	NA	NA
	PFJS impulse	$p < 0.001$	NA	NA

Note: NS, not significant; NA, not applicable.